DTI-LM: Language Model Powered Drug-Target Interaction Prediction

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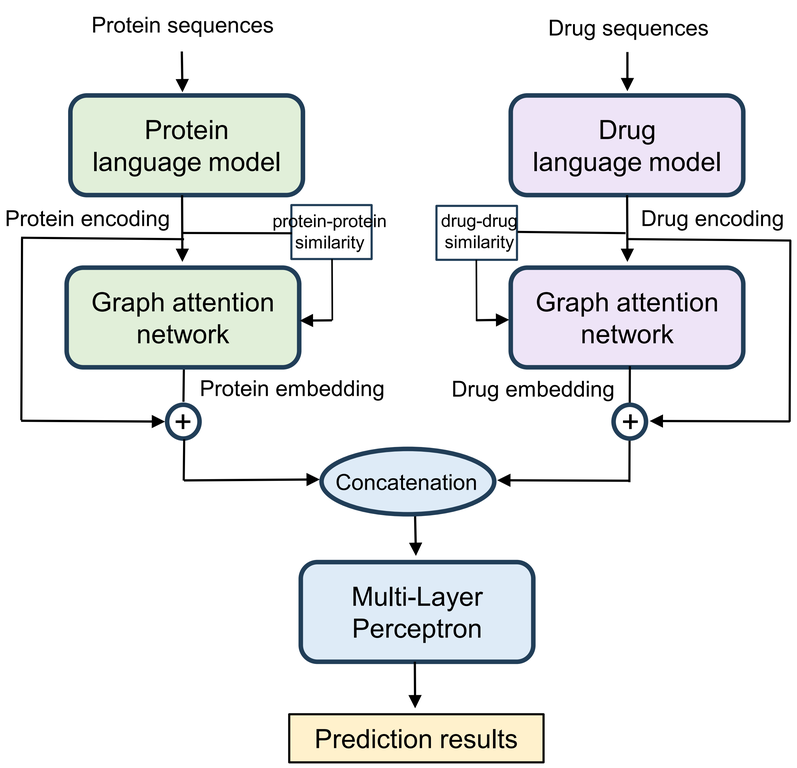
Background

Drug-target interaction (DTI) facilitates novel drug identification and drug repurposing for therapeutic applications. Machine learning-based approaches have shown great promise for DTI prediction. These approaches differ on the datasets they use as input which subsequently dictates their varied architecture. Knowledge graph (KG) based frameworks have shown high prediction accuracy particularly for cold start problems. In cold start problems, drugs or proteins are exclusive in train and test set. However, these frameworks require significant volume of data and computational resources. On the other hand, 1D sequences (protein amino acid, drug molecular description or SMILES) are most easily available and enables faster processing but are vulnerable to cold start problems.

Objective

- We propose DTI-LM, a language model-based DTI prediction framework that provides accurate DTI prediction with fraction of required data and computational resources compared to KG based approaches.
- This model uses 1D sequences as input to provide comparable results to state-of-the-art literature.
- We further investigate the current limitations of language models for DTI prediction and suggest possible paths to improve the prediction performance.

Method and Research Design



Results

1D sequence-based model comparison:

		DTI-LM	TransDTI	DeepDTA	DeepDTI
balanced	random	**0.951**	0.934	0.889	0.916
		0.953	0.935	0.882	0.914
	cold start	**0.902**	0.877	0.874	0.859
	for drug	**0.899**	0.889	0.871	0.868
	cold start	**0.923**	0.916	0.855	0.838
	for protein	**0.935**	0.920	0.825	0.850
unbalanced	random	**0.960**	0.952	0.907	0.947
		0.863	0.858	0.623	0.773
	cold start	**0.890**	0.876	0.765	0.860
	for drug	**0.674**	0.651	0.441	0.582
	cold start	**0.938**	0.916	0.737	0.871
	for protein	**0.821**	0.789	0.441	0.614

Table 1. **The classification performance on DrugBank dataset.** Average AUROC and AUPRC scores of drug-target interaction prediction for random, cold start for drug, and cold start for protein data splitting.

		DTI-LM	TransDTI	DeepDTA	DeepDTI
balanced	random	**0.939**	0.926	0.868	0.923
		0.934	0.918	0.729	0.910
	cold start	**0.872**	0.870	0.754	0.863
	for drug	0.879	0.878	0.699	**0.886**
	cold start	**0.812**	0.809	0.697	0.757
	for protein	**0.787**	0.779	0.572	0.767
unbalanced	random	**0.945**	0.941	0.820	0.935
		0.839	0.834	0.577	0.813
	cold start	0.895	0.872	0.851	**0.896**
	for drug	**0.744**	0.708	0.637	0.743
	cold start	0.831	0.818	**0.869**	0.761
	for protein	0.463	0.456	**0.568**	0.366

Table 2. **The classification performance on BindingDB dataset.** Average AUROC and AUPRC scores of drug-target interaction prediction for random, cold start for drug, and cold start for protein data splitting.

Table 1-4 reports the average prediction results for different prediction scenarios. First rows corresponding to each splitting strategy represents the AUROC and the second row represents AUPRCs scores. Table 1 and 2 compares the 1D sequence-based models. Table 3 and 4 compares both 1D sequence-based models and KG based models (DTiGEMS+[2], TriModel[3], DTINet[4], KGE_NFM[1])

Overall comparison:

		DTI-LM	TransDTI	DeepDTI	MPNN_CNN	DTiGEMS+	TriModel	KGE_NFM
balanced	random	**0.974**	0.969	0.865	0.834	0.964	0.951	0.968
		0.966	0.961	0.820	0.788	0.957	0.946	0.961
unbalanced	random	0.984	0.984	0.982	0.974	0.976	**0.985**	0.983
		0.930	0.927	0.917	0.874	0.874	0.886	0.902
	cold start	0.785	0.762	0.628	0.629	0.745	0.817	**0.853**
	for drug	0.451	0.442	0.191	0.194	0.518	0.503	**0.521**
	cold start	0.911	0.902	0.497	0.502	0.674	0.829	**0.921**
	for protein	**0.739**	0.729	0.099	0.098	0.443	0.483	0.679

Table 3. **The classification performance on Yamanishi_08 dataset.** Average AUROC and AUPRC scores of drug-target interaction prediction for random, cold start for drug, and cold start for protein splitting. DeepDTI, MPNN_CNN, DTiGEMS+, TriModel, and KGE_NFM results are directly reproduced from Ye et al.[1]

		DTI-LM	TransDTI	DeepDTI	MPNN_CNN	DTINet	KGE_NFM
balanced	random	**0.944**	0.938	0.859	0.830	0.940	0.903
		0.948	0.939	0.840	0.805	0.941	0.898
unbalanced	random	**0.971**	**0.971**	0.952	0.929	0.944	0.962
		0.906	0.902	0.793	0.705	0.817	0.855
	cold start	0.760	0.742	0.662	0.806	0.853	**0.881**
	for drug	0.393	0.383	0.225	0.462	**0.592**	0.555
	cold start	**0.832**	0.823	0.487	0.431	0.778	0.813
	for protein	**0.595**	0.589	0.092	0.078	0.388	0.444

Table 4. **The classification performance on Luo's dataset.** Average AUROC and AUPRC scores of drug-target interaction prediction for random, cold start for drug, and cold start for protein splitting. DeepDTI, MPNN_CNN, DTINet, and KGE_NFM results are directly reproduced from Ye et al.[1]

Language model (LM) encoding analysis:

	raw sequences		LM encoding	
dataset	drug	protein	drug	protein
DrugBank	0.101	0.072	0.644	0.853
BindingDB	0.117	0.090	0.574	0.859
Yamansihi_08	0.104	0.089	0.554	0.853
Luo's dataset	0.097	0.078	0.488	0.845

Table 5. **LM encoding similarity.** Similarity is measured on the raw sequences and language model encoding representing drugs and proteins.

	raw sequences		LM encoding	
dataset	drug	protein	drug	protein
DrugBank	25.1%	0.0%	14.3%	30.7%
Yamansihi_08	14.1%	21.5%	6.5%	44.0%
Luo's dataset	30.9%	0.0%	24.5%	26.4%

Table 6. **Top 5 neighbor support.** Average percentage of interactions shared by majority of the neighbors.

Table 5 reports sequence and encoding similarity for drugs and proteins. Table 6 reports the percentage of interactions shared by majority of top 5 neighbors selected based on highest similarity measured in Table 5.

Conclusions

- DTI-LM outperforms the baselines consistently on random splitting, and solved the problems associated with cold start for protein prediction.
- DTI-LM although improved over the results of other 1D sequence-based approaches for cold start of drugs predictions, it falls behind KG based approaches.
- Analysis shows the embedding generated by the chemical language model is not as efficient as the protein language model to find similarity between drugs.
- Without capturing drug similarity, test set of cold start for drug prediction will contain unknown drug embeddings, therefore worse results.
- Current chemical language models are smaller which might affect its performance and a better language model may alleviate the challenge.
- The difference in the two language models also explain the difference between the improvements in cold start for protein and cold start for drug predictions.

References

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